Embedded Memory Design Project Report

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Circuit Design and Simulation

Single ADC column

A single ADC column comprises 32 rows of compute cells. Each compute cell includes:

- A 9T SRAM cell for 1-bit weight storage
- A transmission gate enabling multiplication operation
- A 1 fF capacitor for charge accumulation
- A 2-bit DAC that generates the input analog voltage

During operation, the weighted analog contributions from all 32 rows are accumulated through charge sharing across the capacitors, producing an averaged MAC voltage at the column output.

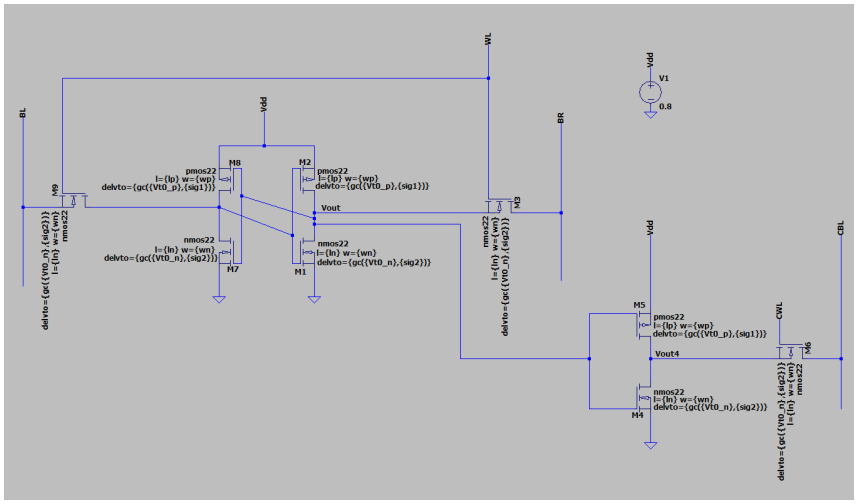
The output is passed on to the ADC through buffer and additional circuitry for sampling.

Compute cell

We have used a 9T cell which comprises of a conventional 6T SRAM cell and an additional 3T for disturbance isolation. The 3T consists of an inverter and an NMOS access transistor to isolate the storage nodes from switching noise on CBL(Compute Bit Line). This could also enable switching between different voltage rails.

Length:
Length of 22nm and 44nm for all transistors were compared. With 22nm, read margin was close to 0, therefore eliminating the need to check write margin.

Width:
Reducing width increases write margin. It is kept at an optimal value to keep the area of the cell low and having a reasonable write margin at the same time.

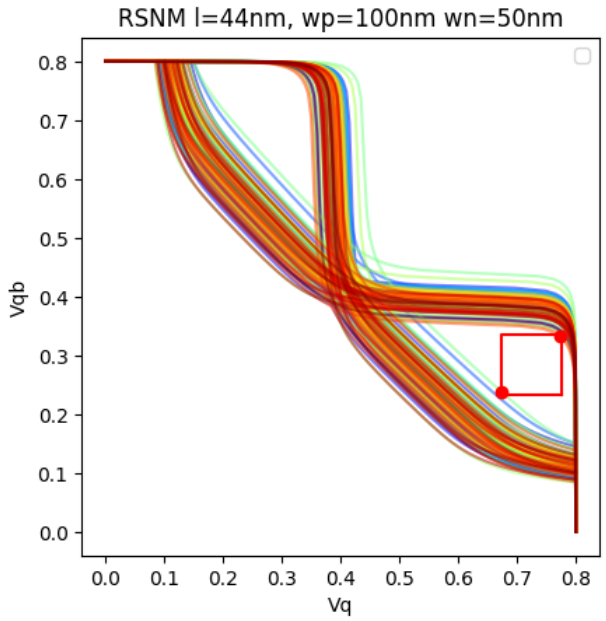


Sizing

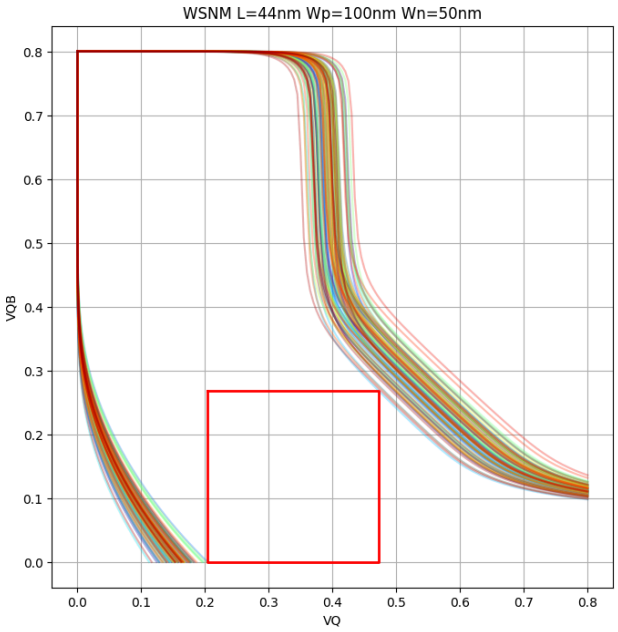
Cell sizing: (all in nm)
PMOS: (44*100)*3=13200 nm^2
NMOS: (44*50)*6=13200 nm^2

Total area per compute cell = 26400 nm^2

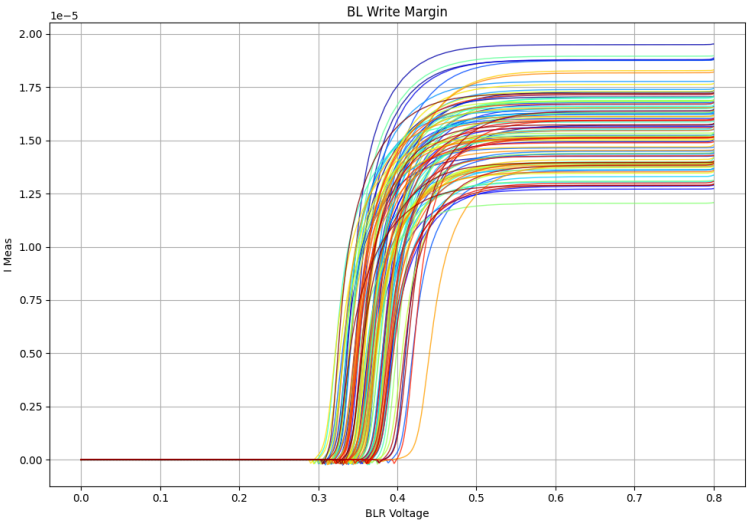
Margins



RSNM = 0.153V



WSNM = 0.379V



BLWM = 0.3V (approx)

Functionality

A 1 bit weight is stored in the 9T cell. When CWL(Compute Word Line) is driven high, the stored weight appears on the CBL(Compute BitLine).

- If the CBL is driven high, the transmission gate turns ON, allowing the input analog voltage to be transferred to the top plate of the accumulation capacitor.
- If the CBL is low, the transmission gate remains OFF and the capacitor is discharged.

The input voltage is generated by a DAC driven by a 2-bit digital input.

The charge accumulated on the 32 capacitors undergoes charge sharing resulting in an averaged analog MAC voltage.

Digital-to-Analog Converter

Architecture and sizing

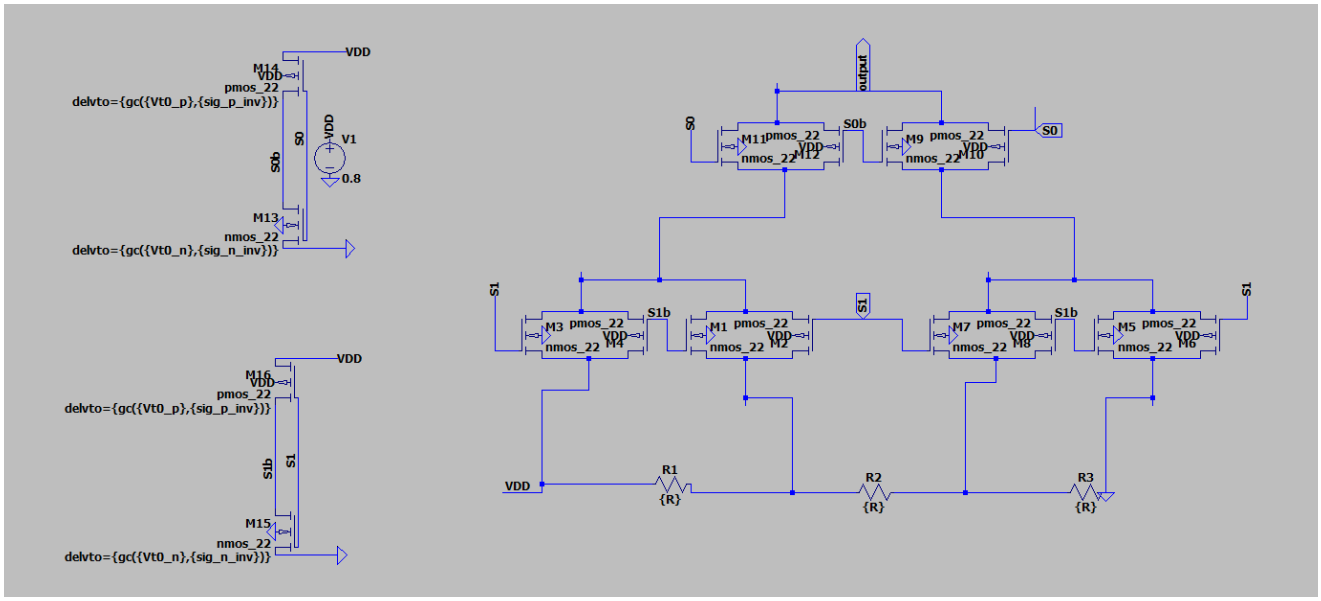
A resistor string DAC with equal resistances of 1kOhm is used with a 2 stage 4:1 Mux. The resistor string acts as a potential divider, giving 4 reference voltages. 0, ⅓ VDD, ⅔ VDD and VDD.

2 bit digital input is given as S0 and S1.

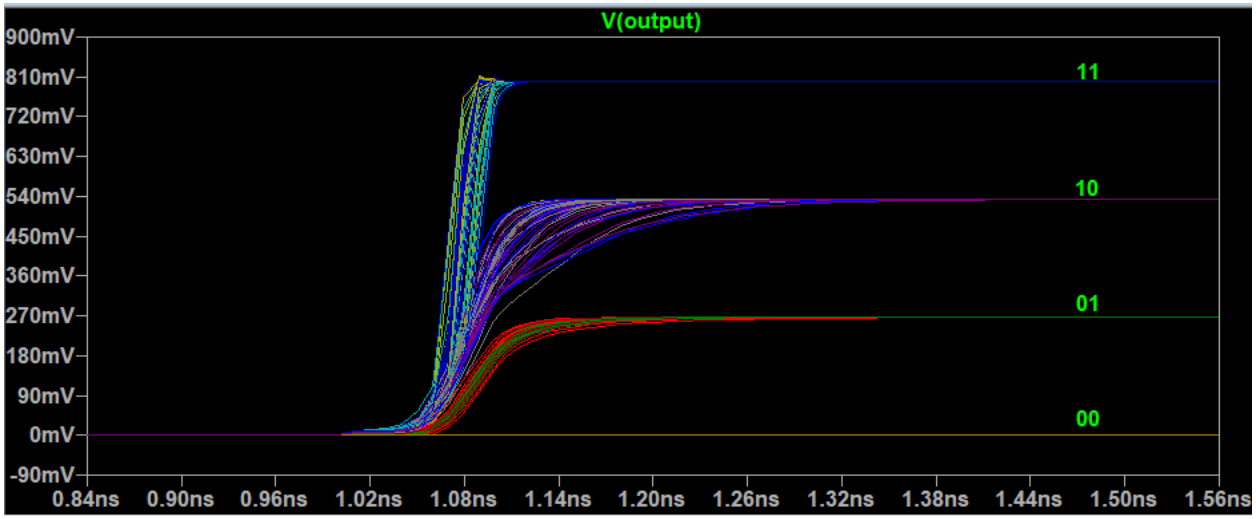
NMOS: (22*50)*8=8800 nm^2
PMOS: (22*100)*8=17600 nm^2
Resistor:
Assume we use Rpoly with sheet resistance 500 Ohm/square (typical value for 22nm technology)
W=0.5um L=1um
Area=0.5 um^2

Total area=0.526 um^2

Schematic



Voltage output vs time for the various input codes

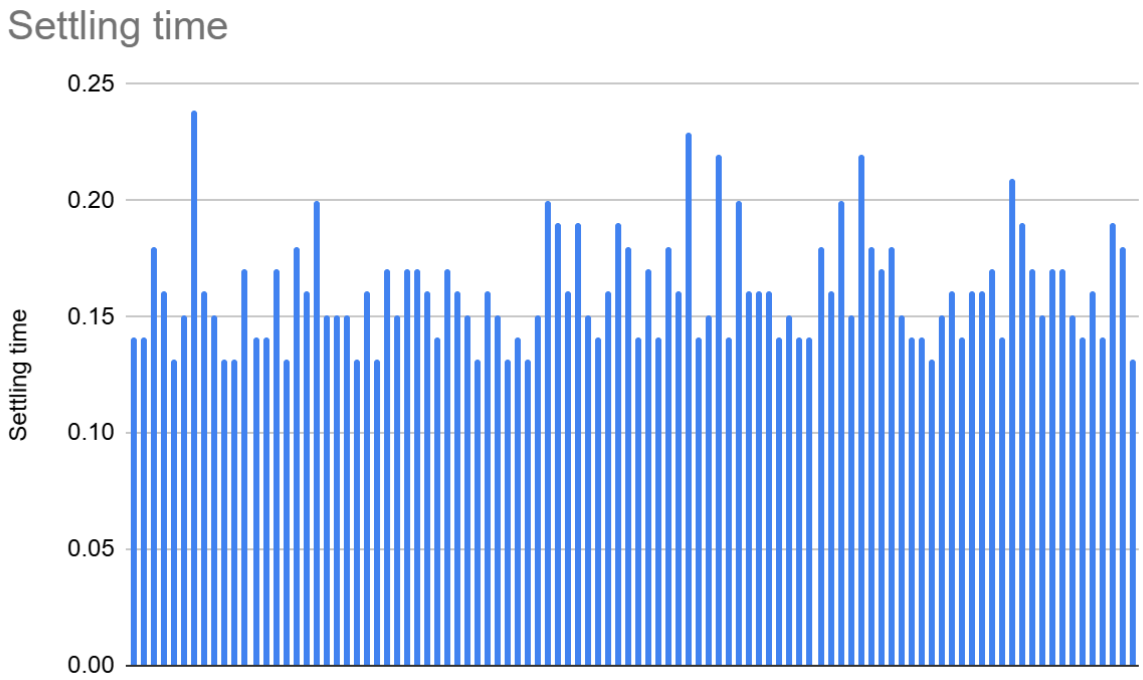


Digital Code	Voltage Output (mV)
00	0
01	266
10	533
11	800

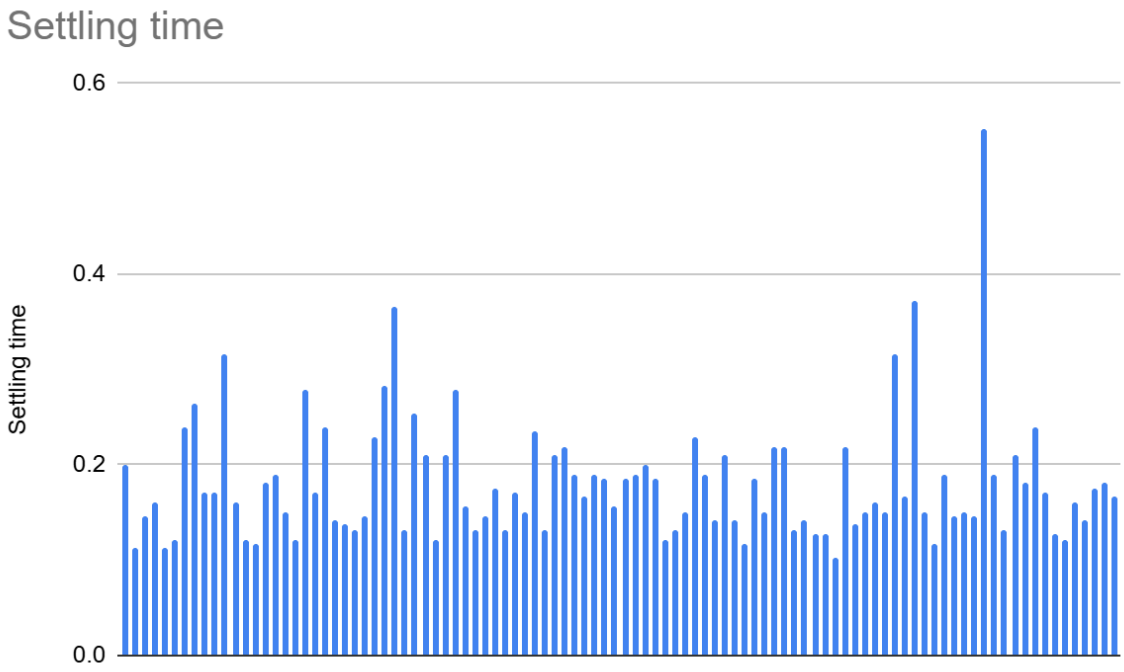
Settling time for each input code

1) DAC input = 00
   Settling time = 0ns

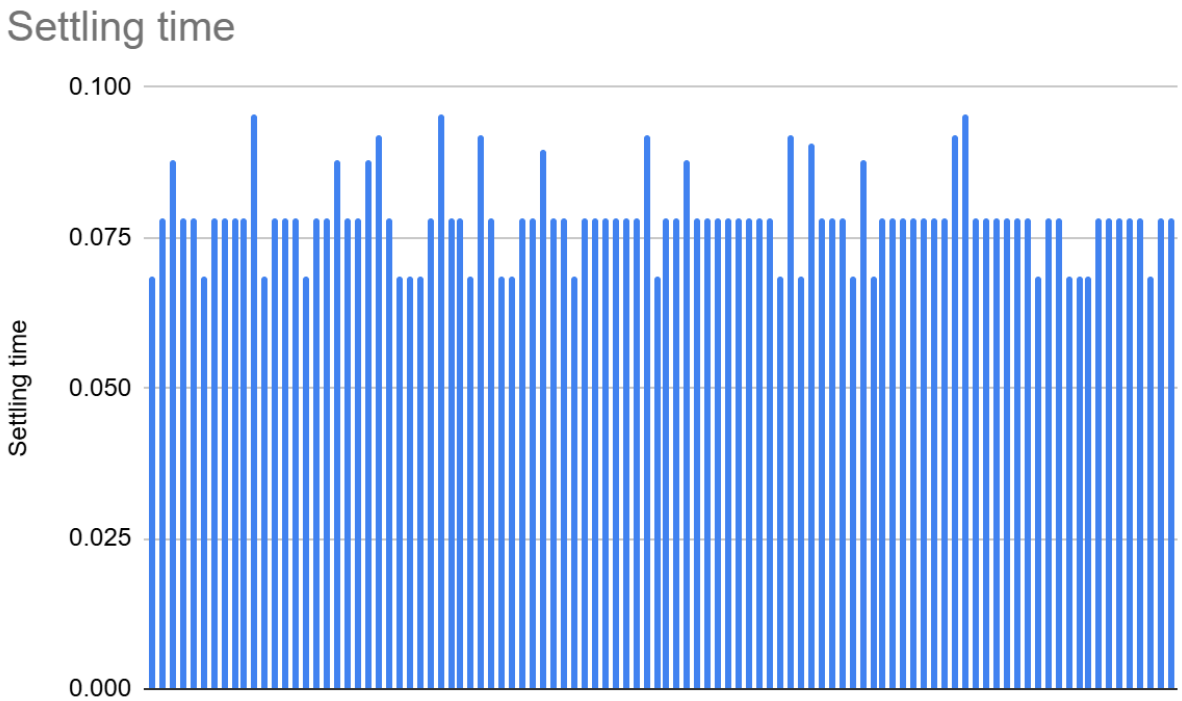
2) DAC input = 01
   Maximum settling time = 0.2388 ns



3) DAC input = 10
   Maximum settling time = 0.55 ns



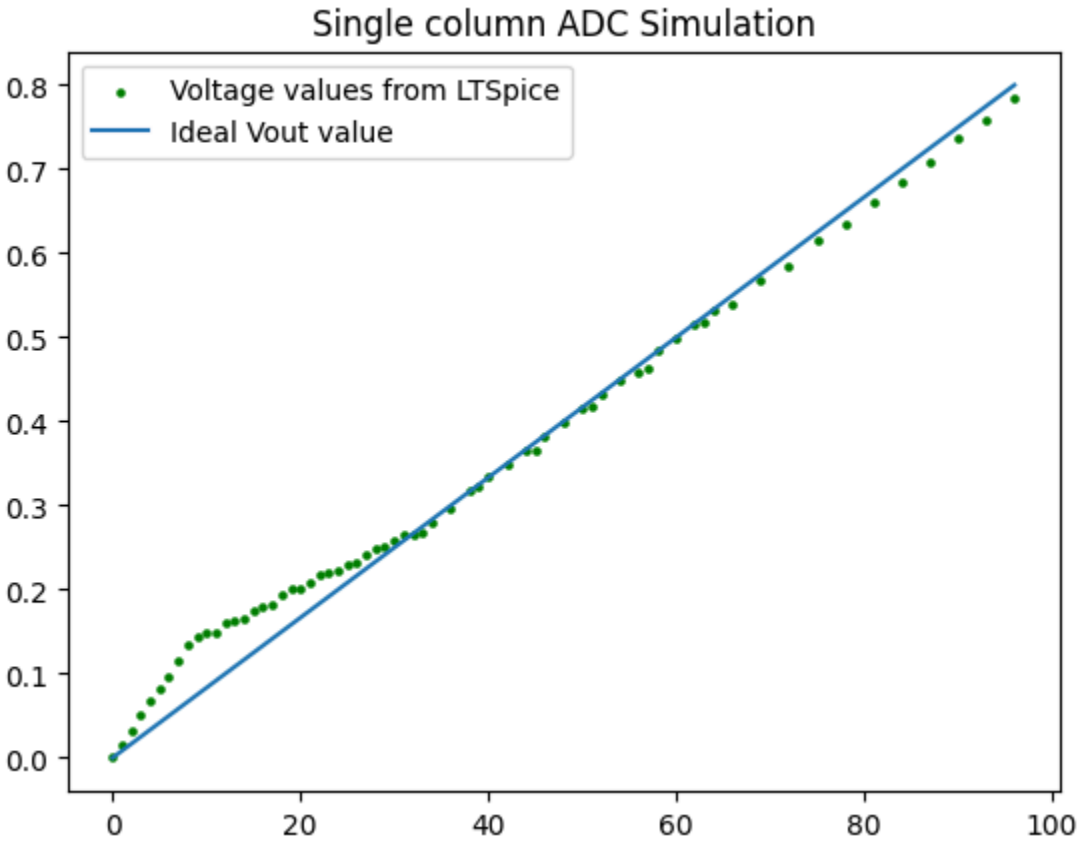
4) DAC input = 11
   Maximum settling time = 0.0955 ns

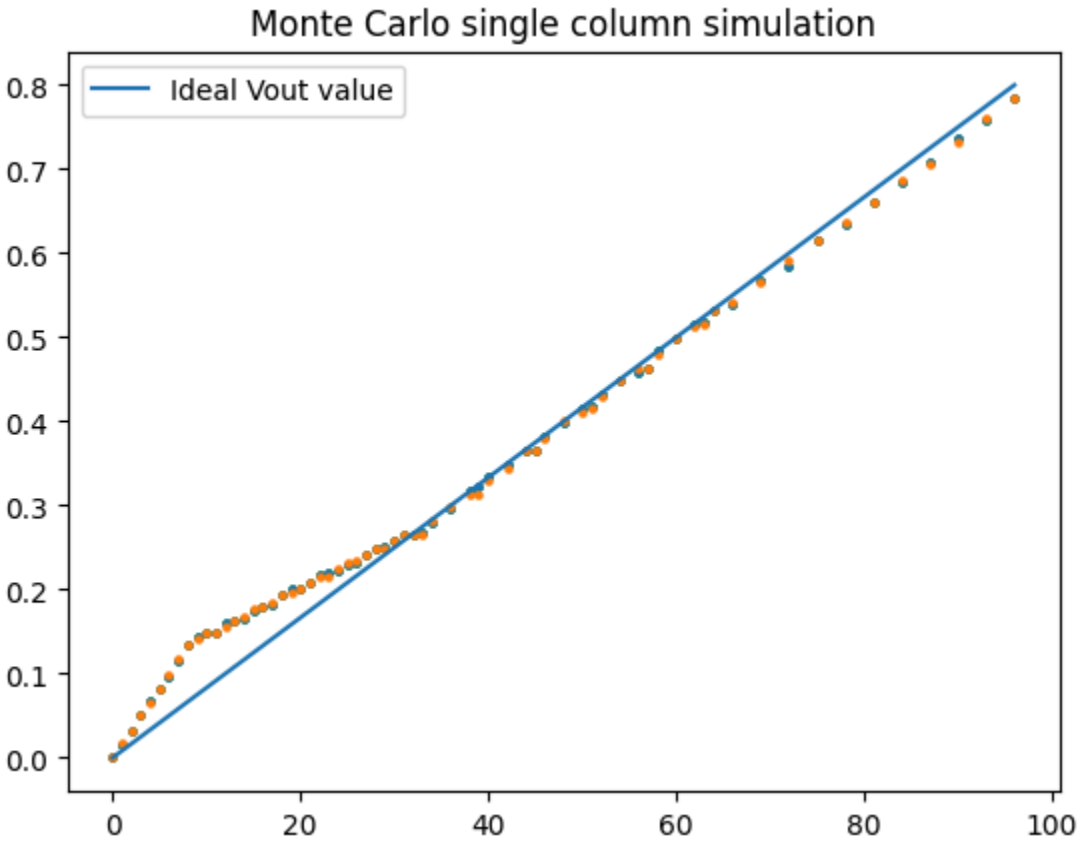


Limitations of architecture

- Since it is a 2 stage mux, high transistor widths would be required since the transistors are in parallel.
- This would add up to high parasitic capacitances and resistances at the output node.
- The output node drives the reference voltage for multiplication operation. Hence this may introduce noise in the accumulation.

Single column simulation to demonstrate synthetic MAC range

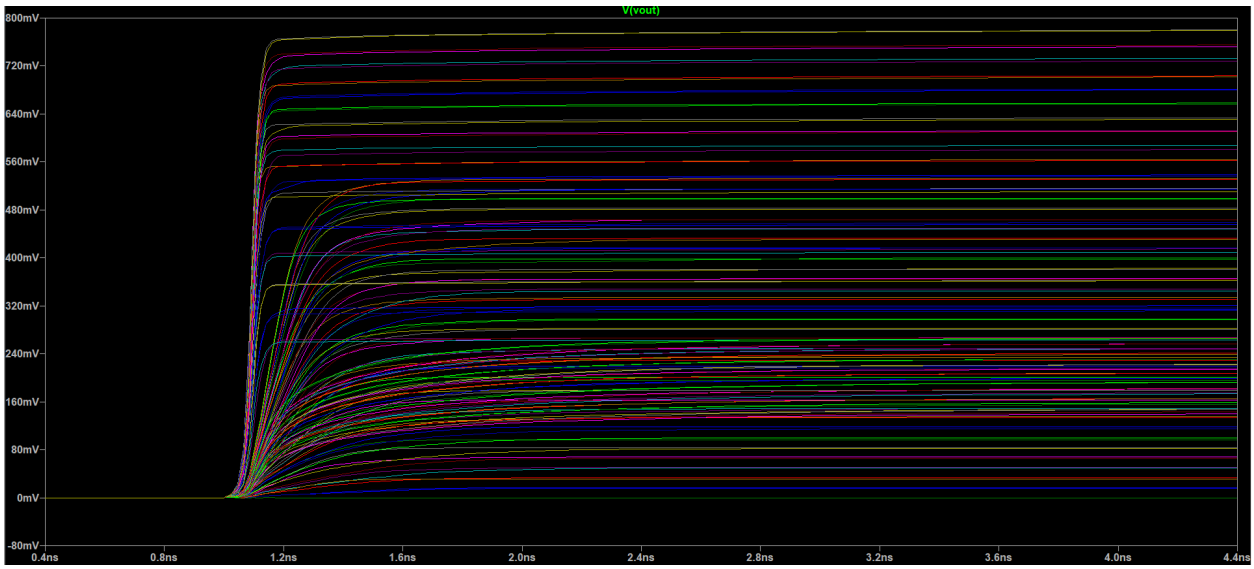


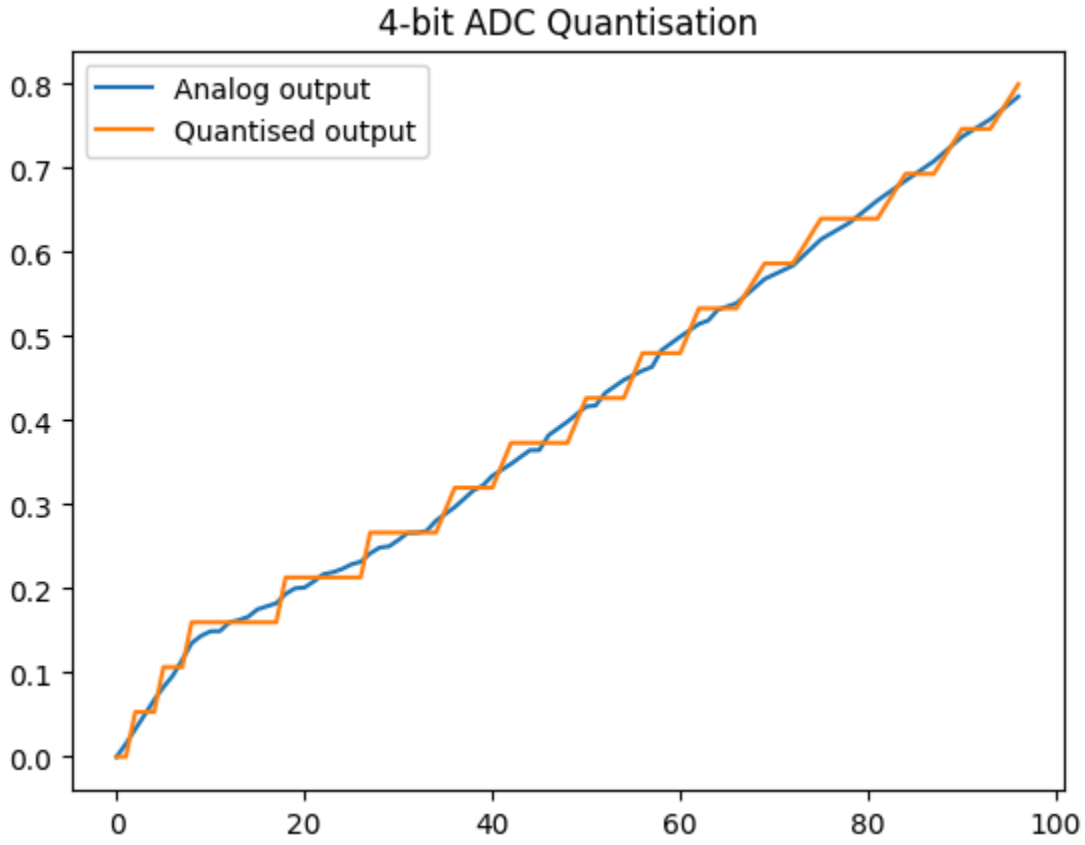


Our MAC ranges from 0 to 96. The measured voltage output is linear for MACs 30 - 96. However, there is significant non-linearity for lower MAC values, possibly due to coupling from transmission gate transistors.

From the error in the observed values, the standard deviation of voltage is 0.025. This error is incorporated into the noise in the MATLAB code to obtain accuracy with process variations.

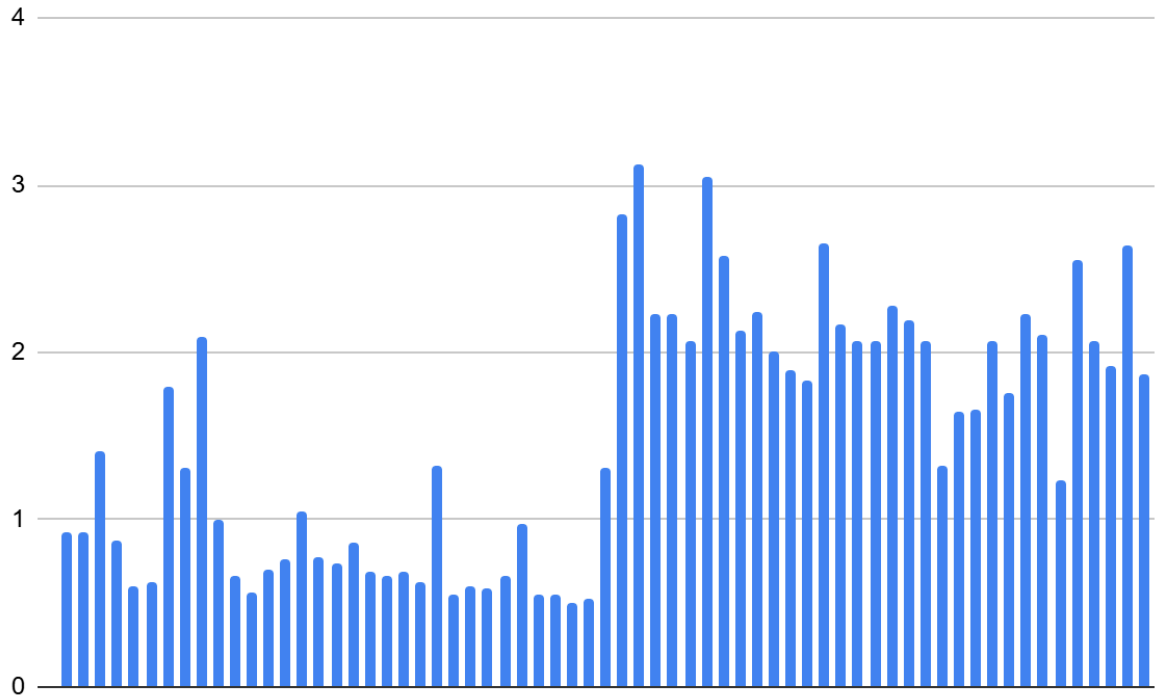
Bitline voltage for various input codes in one Monte Carlo run





T_{MAC} and I_{MAC} estimate

Settling time for MAC voltage on the bitline



Max settling time =3.12ns

Therefore T_{MAC}=3.12 ns

Each column performs 128 1-b operations. This gives 40 1-b GOPS. Our architecture uses tiles of 32 x 120 cells, giving 4.8 TOPS per tile.

T_{MAC}	3.12ns
Number of 1-b operations per column	128
Number of 1-b operations per tile	15360
Number of 1-b operations per macro	322560
TOPS per tile	4.9 TOPS
TOPS per macro	102.9 TOPS

Current values:

Compute cell and switch:
Steady value = 280nA
Peak value = 1.36uA

DAC:
300nA peak
200pA steady value

There are 32 rows per column for 1 MAC operation.
I_{MAC} = 32*280 nA per column
=8.96 nA
I_{PEAK} =32*1.36 uA
=43.52 uA

Total current consumption = 15000*8.96 =134400 nA = 0.134 mA

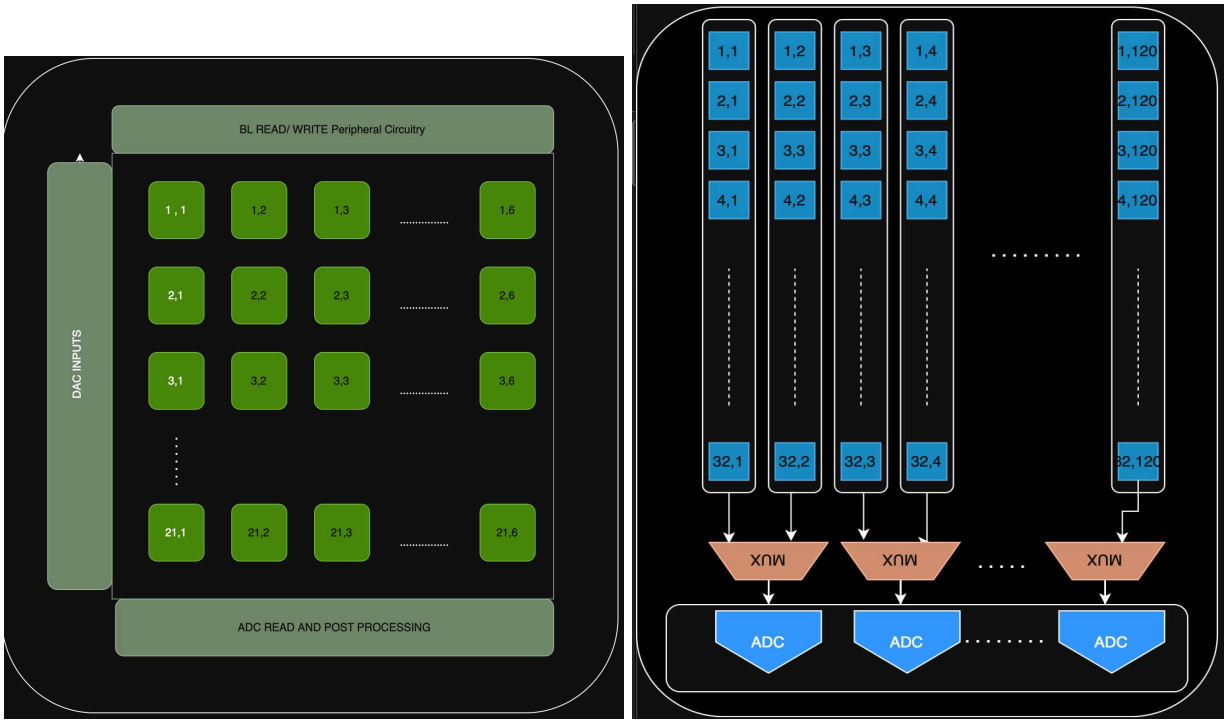
I_{MAC}=0.134 mA

Memory Organisation

We need 784 x 100 for MNIST
Each weight corresponding 6 bits so **784 x 100 x 6 = 470400**
Our Mac unit has 32 x 120 which is 20 neurons partial sum where each column will have 2 bit input through DAC and we get 32 rows accumulation in each of these bitlines. Now we can mux the 4 bit ADC depending on the size of the ADC to each of the columns and we can shift and add these Mac outputs later accordingly.

To accommodate for negative weights we are using two different banks where negative and positive weights are separated and after respective mac accumulation, they will be digitally subtracted to get correct output. This ensures generality of the MAC unit so that it can work for non negative weight models also.

Using 32 x 120 as 1 tile, we can use 126 such tiles as 21 x 6 dimension which gives nearly a total number of bits as **32 x 120 x 126= 483840** which gives us an aspect ratio of **1 : 1.07** approximately.



The figure above corresponds to the entire architecture followed by each tile overview having 32 x 120 cells in each tile. Muxing of ADC is just for an idea for image, actual value depends on size of ADC and other stuff so please not to scale.

MATLAB DELIVERABLES FOR PHASE 2
Introducing Non linearity in the architecture through Matlab.

The selected network parameters were:

B_x = 2, B_w = 6, B_{ADC} = 4.
Monte-Carlo simulations of the full ADC column showed that the worst-case variation metric was:
$\sigma/\Delta(step)$ = 0.025,
$\Delta(step)$ = R / 2^(BADC)
where:
• σ is the standard deviation of the analog output distribution,
• $\Delta step$ is the mean ADC step

Effective variance is the sum of variances as they are correlated. So Enob is measured as

$$ENOB = B_{ADC} - \log_2 \sqrt{1 + 12(\frac{\sigma}{\Delta_{\text{step}}})^2}$$

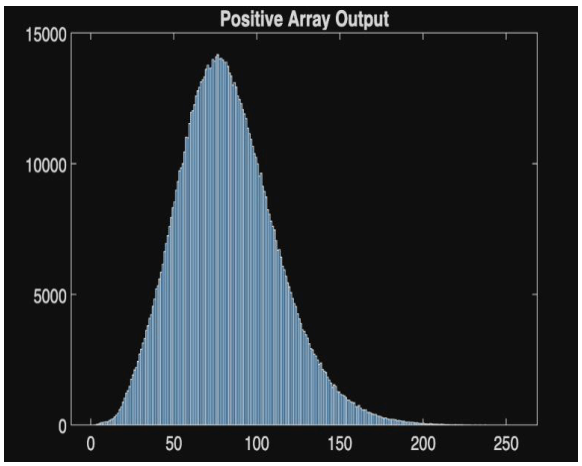
Using this the calculated ENOB turned out to be **3 bits**. Degradation of 1 bit precision from ADC.

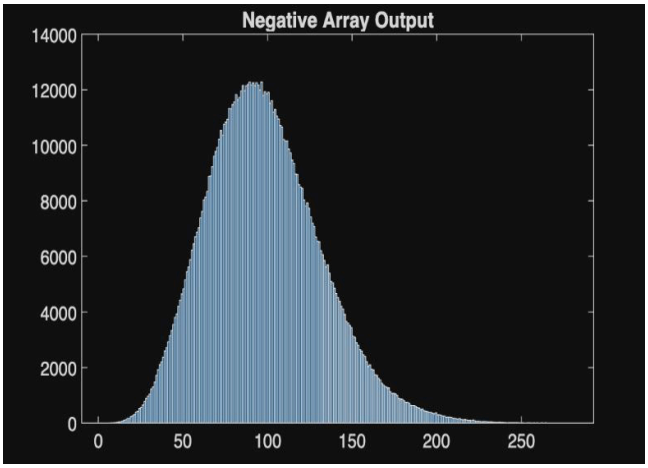
Using this ENOB the matlab accuracy is 95.66% and the drop is **0.19%.**

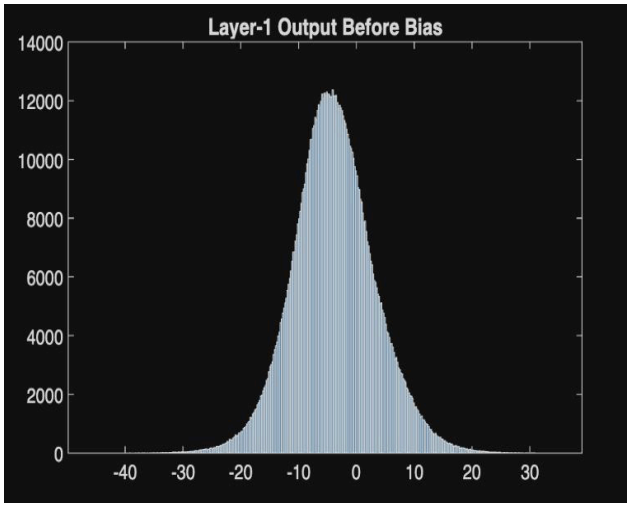
The small degradation in accuracy demonstrates that the proposed charge-domain CIM architecture is robust against realistic analog non-idealities and process variation.

(The code will be submitted with the report.)

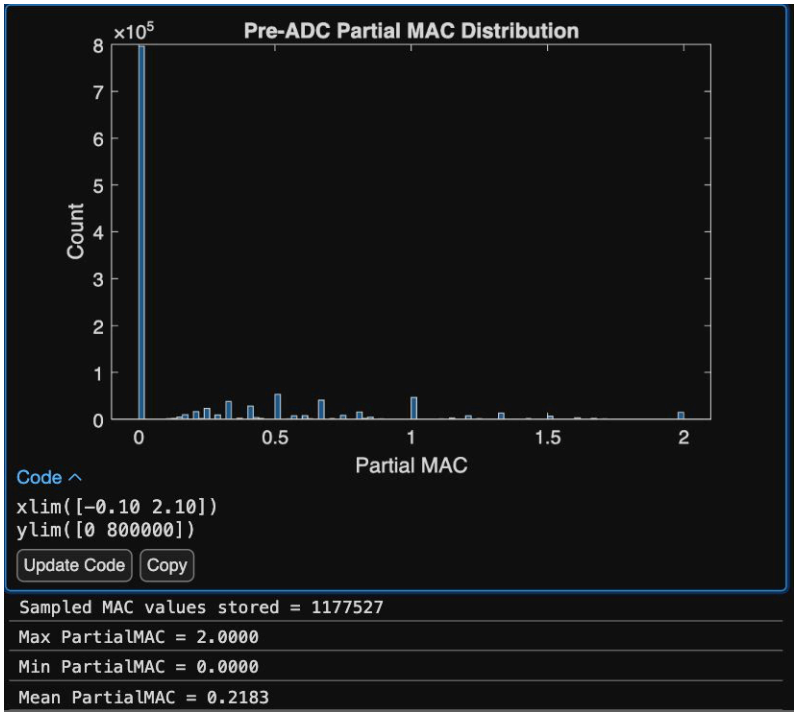
Extra matlab deliverables: Some from phase 1







The above are the histogram of respective positive and negative weight sums which are digitally subtracted later to get the final mac histogram as above.



This is the histogram of partial mac sums in each mac unit for a given data set. This is to show that my range of values lie inside my ADC full precision which is 3.